电路基础小测：

1245全部正确，3的改正如下：

1. $i_2 = i_1 = 2A$，$P_1 = -u_1 i_1 = \boxed{-20W}$

KCL：$i_2 + i_3 + i_4 = 0 \Rightarrow i_4 = -i_2 - i_3 = -2 + 3A = 1A$，$P_4 = u_4 i_4 = \boxed{5W}$

$u_3 = -u_4 = -5V$，$P_3 = -u_3 i_3 = -(-5V) \times (-3A) = \boxed{-15W}$

KVL：顺时针：$u_1 - u_3 - u_2 = 0$，$u_2 = u_1 - u_3 = 10V - (-5V) = 15V$

$P_2 = u_2 i_2 = 15V \times 2A = \boxed{30W}$

2. 节点电压法：③接地，设 u_{n1}、u_{n2}

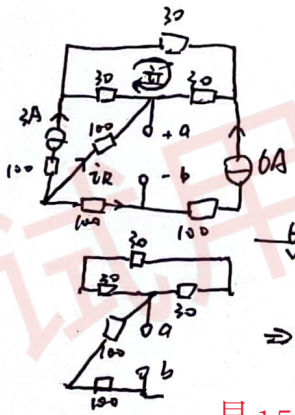
得到：$\begin{cases} 4A + u_{n1} \times 1S + (u_{n1} - u_{n2}) \times 2S = 0 \\ -24\cos\omega t + u_{n2} \times 2S + (u_{n2} - u_{n1}) \times 2S = 0 \end{cases} \Rightarrow \begin{cases} u_{n1} = (6\cos\omega t - 2)V \\ u_{n2} = (9\cos\omega t - 1)V \end{cases}$

故 $u_1 = u_{n1} = (6\cos\omega t - 2)V$

$u_2 = u_{n2} = (9\cos\omega t - 1)V$

$u_3 = u_{n1} - u_{n2} = (-3\cos\omega t - 1)V$

3. (1)戴维南定理：开路电压 ⇒



~~设一个回路电流 i_1 ⇒ $i_1 \times 90\Omega - 3A \times 30\Omega + 6A \times 30\Omega = 0$~~

~~$\Rightarrow i_1 = -1A$~~

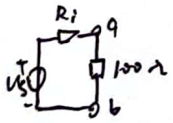
$3A + 6A + i_R = 0$，$i_R = -9A$

~~$U_S = U_{ab} = -i_R \times 100\Omega$~~ $u_{ab} - 6A \times 100\Omega + i_R \cdot 100\Omega = 0$，$U_S = U_{ab} = 1500V$

电流源 → 开路 计算 R_i

$\Rightarrow R_i = 200\Omega$

等效为



$i = \dfrac{U_S}{R_i + R_{ab}} = \dfrac{900}{200 + 100} A = 3A$

是1500V，抄错了，故结果是5A

(2)诺顿定理：短路电流 ⇒



$-i_1 \times 90\Omega - 3A \times 30\Omega + 6A \times 30\Omega = 0 \Rightarrow i_1 = 1A$

$i_2 \times 200\Omega - 6A \times 100\Omega + 3A \times 100\Omega = 0 \Rightarrow i_2 = 1.5A$

$\Rightarrow I_S = 6A - i_2 = 4.5A$

等效为



$R_i = 200\Omega$，$G_i = \frac{1}{200}\Omega^{-1} \Rightarrow i = I_S \times \dfrac{R_i}{R_i + R_{ab}} = 4.5 \times \dfrac{200}{200+100} = 3A$

4. $3H$ 并 $6H = \dfrac{3 \times 6}{3+6} H = 2H$

$3F$ 并 $6F = 9F$

故 $L = \dfrac{3 \times 6}{3+6} + 3H = 5H$

$C = \dfrac{3 \times (3+6)}{3 + (3+6)} C = 2.25F$

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5. $Z = (j10\Omega \,//\, -j20\Omega) + 10\Omega = \dfrac{200}{-j10} + 10\ \Omega = (10 + j20)\Omega$

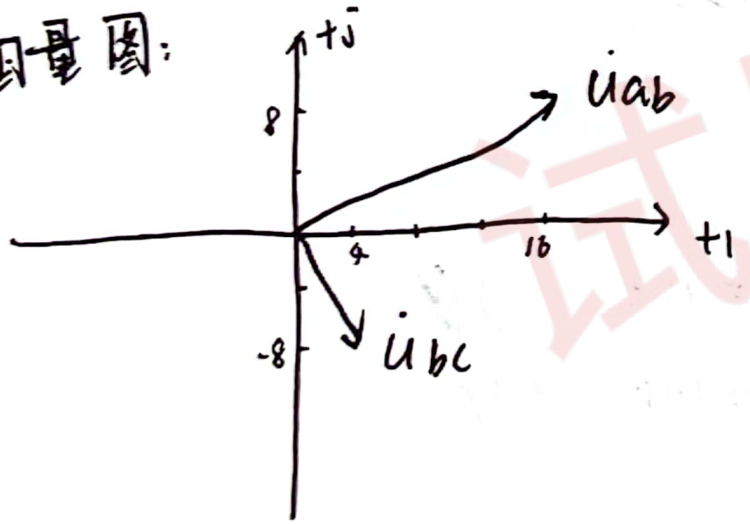
$\dot{U} = 20\angle 0^\circ V = 20V$

$\dot{I} = \dfrac{\dot{U}}{Z} = \dfrac{20}{10+j20}\ A = (0.4 - j0.8)\ A$

$\dot{U}_{bc} = \dot{I} \cdot R_{bc} = (4 - j8)\ V$

$\dot{U}_{ab} = \dot{U} - \dot{U}_{bc} = (16 + j8)\ V$

相量图：



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